Roll No.: _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

Amrita Vishwa Vidyapeetham, ASPS, Amritapuri

Midterm Examination – March 2025

Second Semester

Integrated MSc Physics/BSc(H) Physics with Minor in Scientific Computing/Data Science

Integrated MSc Maths /BSc(H) with Minor in Data Science

24PHY113 Electricity and Magnetism

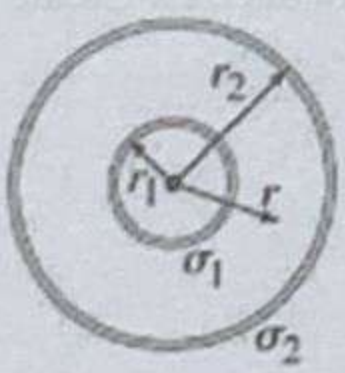
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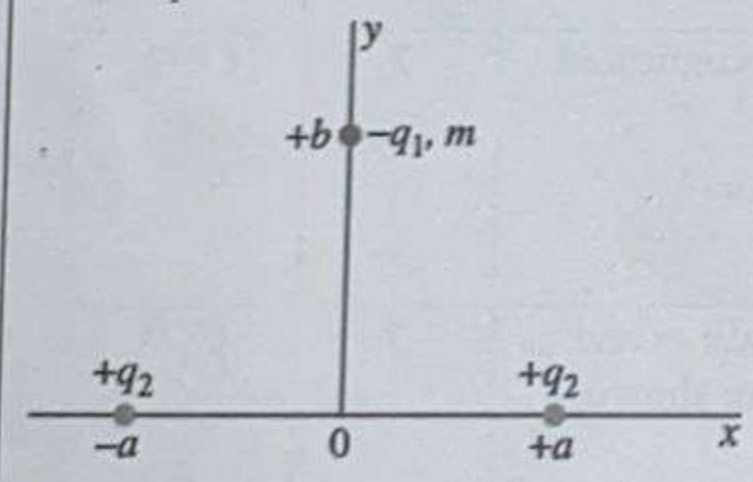
Time :2 hours

Maximum Marks:50

Instructions

1. Answer all the questions.
2. Answer with appropriate significant figures and units.

Qn No.		Marks	CO	Blooms Taxonomy level
1	Two unknown positive charges are separated by a fixed distance r. The sum of the two charges is given as Q_T. Assume one of the unknown charges is q. The Coulomb's constant is k. a) What is the force between the charges in terms of q, k, Q_T and r? b) What should be the value of q and the other charge if the force between them is maximum? c) What should be the value of q and the other charge if the force between them is minimum?	7	CO1	L3
2	There are two capacitors, 0.50 μF and 0.80 μF that are connected in series to a 9 V battery. a) What is the potential difference across the capacitors? b) What is the charge on each capacitor?	7	CO2	L3
3	In the figure, two thin concentric spherical shells of radii r_1 and r_2 with uniform charge densities σ_1 and σ_2 respectively, are shown. Neglect the thickness of the shell. 	7	CO1	L3

	a) What is the electric field for $r < r_1$? b) What is the electric field for $r_1 < r < r_2$? c) What is the electric field for $r > r_2$? d) Under what conditions will the electric field be zero for $r > r_2$?			
4	A conducting sphere of radius 16×10^{-2} m is charged to 680 V (relative to $V = 0$ at $r = \infty$). a) What is the surface charge density, σ? b) What is the distance where the potential due to the sphere is 25 V?	7	CO2	L3
5	In a certain region of space, the potential is given as $V = by/(a^2+y^2)$ where a and b are constants. Determine the electric field and its components.	7	CO2	L3
6	A 4.0 C charge loses 4.8 MJ when it travels from one point to another. a) What is the potential difference between the two points? b) The lost energy is used to heat up water from 20 ^{0}C to 100 ^{0}C steam. What is the mass of the water, m? Assume the heat to boil water to to steam is $Q = (m\, c\, \Delta T) + mL$ where c is 4.186 kJ/kg ^{0}C and $L = 2260$ kJ/kg and ΔT is change in temperature.	7	CO2	L3
7	The figure shows a configuration of three-point charges. What is the minimum velocity required for the charge $-q_1$ to escape to infinity along positive y-axis? The initial distance of $-q_1$ is b from the origin while the two equal charges positive q_2 are separated by a distance $2a$.  	8	CO2	L3

L_1-Remember, L_2-Understand, L_3-Apply, L_4-Analyze, L_5-Evaluate, L_6-Create